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A

Power input (USB-C) and regulators

File: lcd_f042_power.kicad_sch

HDMI receiver LT8619C

File: lcd_f042_hdmi.kicad_sch

B

AT070TN92 panel, bias supplies, VCOM, backlight

File: lcd_f042_panel.kicad_sch

STM32F042F6P6 controller and GT915 touch

File: lcd_f042_mcu.kicad_sch

C

AT070TN92 7" 800x480 TTL panel driver: HDMI (LT8619C) + GT915 touch -> USB HID (STM32F042F6P6).

Power: USB-C 5 V (from Raspberry Pi USB port, ~0.7 A max with backlight at 100 %).

Panel power-up order (MCU controlled): DVDD -> RESET -> VGL/AVDD -> VGH -> data -> backlight.

One I2C bus for LT8619C (0x32) and GT915 (0x5D). Panel scan direction: straps R19 (L/R) and R20 (U/D).

J3 pad numbers equal AT070TN92 FPC pin numbers; use a TOP-contact 0.5 mm 50-pin FPC connector.

VCOM (pins 6/46) is adjustable with RV1 (2.55..4.5 V): trim for minimum flicker.

D

LT8619C HDMI->RGB888, TPS61040 bias, TPS61165 backlight, STM32F042F6P6 USB touch

Sheet: /
File: lcd_f042.kicad_sch

Title: AT070TN92 HDMI + USB-touch driver board (STM32F042F6P6)

Size: A4Date: 2026-10-04

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

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